

OrthoCache Pipeline Architecture

Integration with Gemma 4 E2B Hybrid Attention

$$\text{TV}(\alpha, \hat{\alpha}) \leq |S^c| \cdot \exp\left(\frac{\|q\|\sqrt{\varepsilon}}{\sqrt{d_k}} - z_{\max}\right)$$

Lean 4 Verified Truncation Bound



Sliding-Window Layers (12/15)

512-token local cache $\rightarrow$ 1 block
 $\rightarrow$ No eviction needed

Global Attention Layers (3/15)

Full-sequence cache $\rightarrow$ 8 blocks
 $\rightarrow$ OrthoCache target